

# Predictive and Generative AI

Project: Verified Study Buddy

## 1. Your mission

### Your learning target

I can distinguish discriminative prediction from generation by inspecting the requested output.

I can calculate an empirical next-token distribution from counted continuations.

I can separate next-token fit from factual support and downstream usefulness.

### Remember before reading

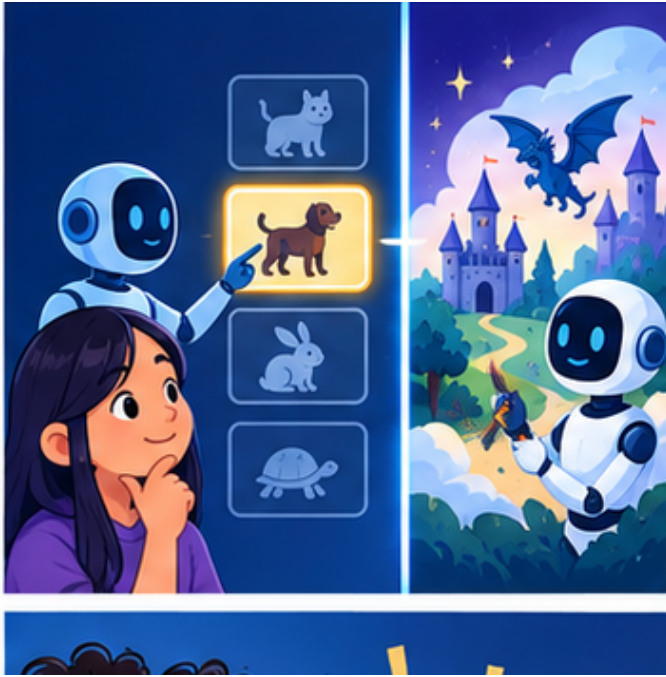
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 1: Predictive and Generative AI

Prediction estimates a target; generation samples a new structured output.



## CREATOR MISSION

Classify systems and explain the output space for each.

Creator name and date:

**Write or draw your reasoning and evidence.**

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## Readiness for the connected explanation

What is the difference between a model prediction and evidence that a statement is true?

## This week's end goal

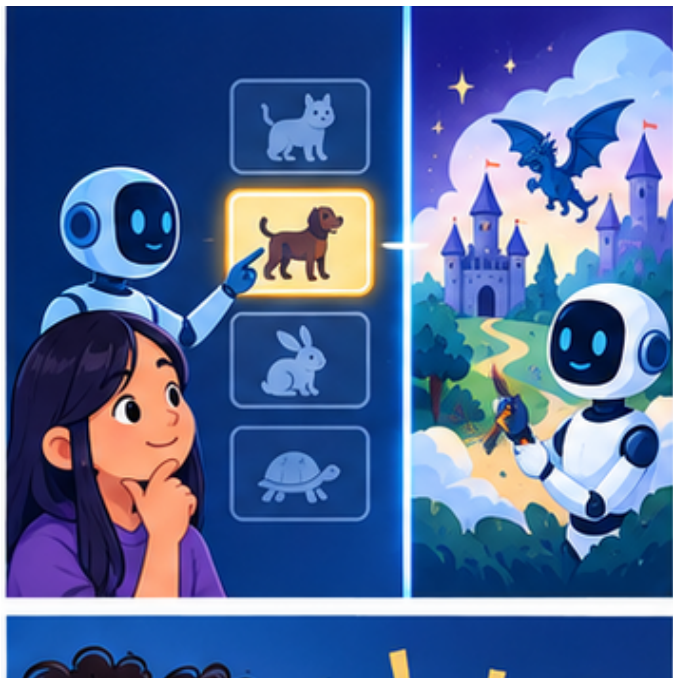
By the end, I can identify generation and prediction objectives without confusing either with verified truth.

# 2. Notice the evidence

## PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use predictive and generative ai to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

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My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

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## A closer explanation

A classifier might map a message to the label request or report. A text generator instead produces a sequence, predicting a distribution over the next token given a prefix. Both make predictions, but the training targets and permitted outputs differ. One application can combine generation with a classifier, retrieval and human approval. Calling a whole product “generative” does not describe every component.

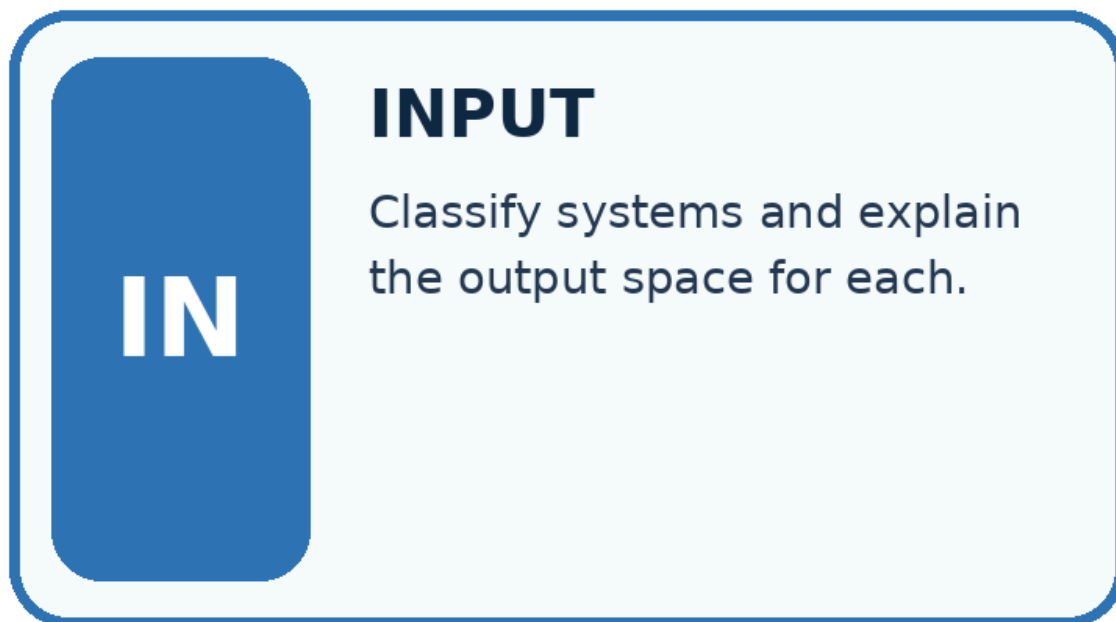
The objective defines what training attempts to fit. Next-token fit rewards matching observed continuations; it does not directly certify that an answer is true, well sourced or permitted. A fluent completion is still an output to evaluate against the task.

## 3. See how it works

### VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## SYSTEM

Prediction estimates a target; generation samples a new structured output.





## EVIDENCE

Classify systems and explain the output space for each.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

### Prefix · Continuation · Count

**Prefix:** The marker is

**Continuation:** red

**Count:** 6

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**Prefix:** The marker is

**Continuation:** blue

**Count:** 3

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**Prefix:** The marker is

**Continuation:** green

**Count:** 1

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There are ten observed continuations. The empirical probabilities are red  $6/10 = 0.6$ , blue  $3/10 = 0.3$ , green  $1/10 = 0.1$ . They are nonnegative and sum to 1. A simple frequentist classroom model uses these relative counts as its next-token distribution. These probabilities describe the supplied corpus counts, not the current colour of any actual marker.

## Keep the evidence roles separate

### Plan or input · Procedure · Permitted conclusion

**Plan or input:** Ten corpus continuations

**Procedure:** Count red 6, blue 3, green 1

**Permitted conclusion:** Distribution 0.6, 0.3, 0.1

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**Plan or input:** Prefix: The marker is

**Procedure:** Select a next token

**Permitted conclusion:** An output, not an observation

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**Plan or input:** Current marker question

**Procedure:** Check direct evidence

**Permitted conclusion:** Colour claim needs support

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The continuation model and a factual observation have different evidence sources.

### Connect the picture and the explanation

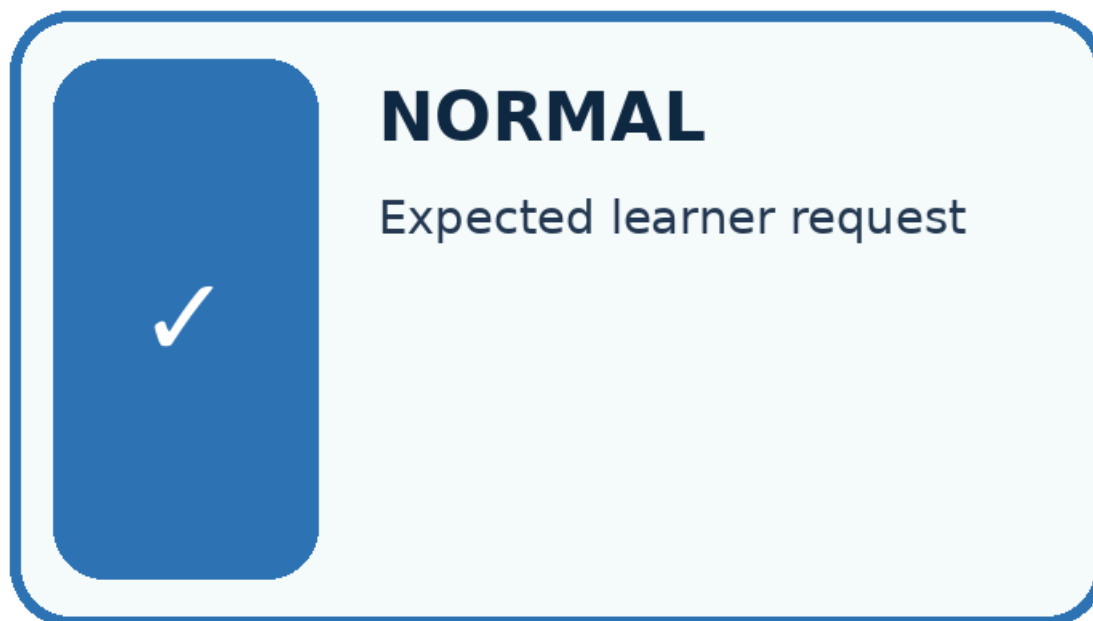
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





## **AMBIGUOUS**

Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



## **MISSING SOURCE**

Evidence unavailable



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: The classifier instead predicts history while N and the draft stay unchanged.

Prediction: The category prediction changes, but the same unsupported draft claim still needs checking.

Reason: Stages can fail independently; one result does not validate another.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

**1**

## **PREDICTIVE AI**

estimates a label, score, or number

**2**

## **GENERATIVE AI**

constructs new content from learned patterns

3

## TARGET

value a predictive system estimates

4

## OUTPUT SPACE

set of possible results

**Word · My own example · Not this / boundary**

**Word:** predictive AI

**My own example:** AI used to estimate a target such as a category, number or future outcome from inputs.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** generative AI

**My own example:** AI that produces new content such as text, images or audio rather than only selecting a

category.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** target

**My own example:** The category, value or outcome a prediction task is intended to estimate.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** output space

**My own example:** The set or range of outputs a system can produce for a task.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

## Words used in the closer explanation

### Word · Meaning

**Word:** Generative objective

**Meaning:** A training aim concerned with producing or modelling possible outputs

\_\_\_\_\_

**Word:** Discriminative objective

**Meaning:** A training aim mapping inputs to a specified target such as a label

\_\_\_\_\_

**Word:** Conditional distribution

**Meaning:** Probabilities for alternatives given stated information

\_\_\_\_\_

**Word:** Autoregressive generation

**Meaning:** Producing a sequence one next item at a time using preceding items

\_\_\_\_\_

## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

### WEEK 1 • WORKED EVIDENCE

- 1 Classifier: choose one label from cat, dog, or rabbit.
- 2 Regressor: estimate a numerical reading time.
- 3 Generator: construct a new explanation token by token.
- 4 A product may combine mechanisms, so test each stage separately.

1. Classifier: choose one label from cat, dog, or rabbit.
2. Regressor: estimate a numerical reading time.
3. Generator: construct a new explanation token by token.
4. A product may combine mechanisms, so test each stage separately.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

After producing red, the next step conditions on the expanded prefix “The marker is red”. A second table might give an end token probability 0.8 and the token today probability 0.2. A generated sequence can stop at the end token or continue. Do not reuse the first-step table unchanged after adding a token unless the model explicitly ignores the new prefix.

A separate classifier can return red as a label from an image; that is a different input and target from continuing text. A downstream evaluation asks whether the system correctly reports the current marker. It must include ground truth or sources and appropriate failure handling, not just count plausible phrases.

## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Predictive and Generative AI

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

### The facts we will use

A fitted classifier labels a question science; its training used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

## Predictive and Generative AI

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Predict category

The classifier predicts science from the new question.



#### 2. Retrieve note

N supplies a specific statement about evaporation.



#### 3. Generate draft

The draft adds "always disappears forever" beyond N.



#### 4. Verify claims

Keep the supported change-of-state claim; remove or review the unsupported addition.

### CHANGE ONE THING

The classifier instead predicts history while N and the draft stay unchanged.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Predict category

**What happens and why:** The classifier predicts science from the new question.

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**Step:** 2. Retrieve note

**What happens and why:** N supplies a specific statement about evaporation.

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### Step: 3. Generate draft

**What happens and why:** The draft adds “always disappears forever” beyond N.

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### Step: 4. Verify claims

**What happens and why:** Keep the supported change-of-state claim; remove or review the unsupported addition.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking. The category being correct does not prove every sentence in the answer is correct.

## What this example does not establish

This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

Read the labelled picture: Identify the first unsupported claim and explain what source would be needed before asserting it.

Change one thing: The classifier instead predicts history while N and the draft stay unchanged. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

### ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



### COMPARE • DIAGNOSE

Weak idea: Every AI system that produces an answer is generative. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

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## A closer explanation

Keep three questions separate: did the model fit continuation data; did the output satisfy the requested format or task; and are factual claims supported? A system that drafts a fictional description has different criteria from one reporting a live measurement. Record the source and time when a claim refers to changing conditions. No AI account is needed to reason about these distinctions.

**Exit: a team says “the highest-probability continuation proves the current fact”. Name the evidence they actually have and the evidence still needed.**

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## 8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



### SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output\_schema LOAD approved\_context or retrieval\_result CALL model or allowed\_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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## A closer explanation

Supported case: the prefix “The door is” is followed in a tiny corpus by open four times and closed once. The empirical probabilities are  $\frac{4}{5}$  and  $\frac{1}{5}$ . A current sensor reports closed. Selecting the most frequent continuation open would be plausible under the corpus model but inconsistent with the sensor observation.

**Guided: compute both probabilities and explain why selecting open does not establish the current door state.**

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## Practise with less help: Compute the distribution

Use workbook W2 for the connected example “Generative and predictive objectives”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Explain the picture

Identify the first unsupported claim and explain what source would be needed before asserting it.

### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



**PREDICT → RUN → VERIFY → EXPLAIN**  
Sort twelve systems by mechanism. For each, name its output space, evidence needed, and one possible failure.

Sort twelve systems by mechanism. For each, name its output space, evidence needed, and one possible failure.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what

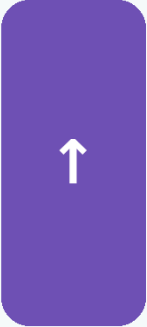
did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

# 10. Build your understanding

## MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

# 11. Connect your project

## CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



**VERIFIED STUDY BUDDY**

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

THIS WEEK’S CHECKPOINT Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

## Evidence for your project

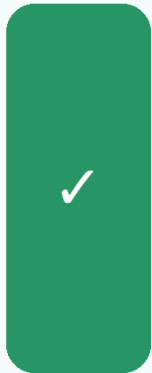
For the Verified Study Buddy, separate routing predictions, source retrieval and generated explanations in its evidence record.

## 12. Show what you know

### INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



### INDEPENDENT CREATOR PROOF

Classify systems and explain the output space for each.

CHALLENGE Classify systems and explain the output space for each.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

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2. My evidence that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

A tool labels a question biology, writes a new question and computes 7 times 8.

#### The question

Identify each operation.

#### Worked explanation

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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## Trace the supplied case visually

### Three outputs need three different checks

#### Evidence or stage · Value or result

**Evidence or stage:** Classify

**Value or result:** Select biology as a category

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**Evidence or stage:** Generate

**Value or result:** Compose a new question

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**Evidence or stage:** Calculate

**Value or result:** 7 times 8 equals 56

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A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

## 14. Try a new case independently

### Independent case: Changed independent case

A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each mechanism and the evidence needed to check its output.

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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## Connect to your project

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

## Transfer practice from the original programme

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

**Explain the mechanism, evidence and one limit.**

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## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

**By the end, I can · How I will show it**

**By the end, I can:** Distinguish discriminative prediction from generation by inspecting the requested output

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Calculate an empirical next-token distribution from counted continuations

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Separate next-token fit from factual support and downstream usefulness

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Distinguish discriminative prediction from generation by inspecting the requested output

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.